| EXPT : 4 | EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION |
|---|---|
| DATE : | |

## AIM

To perform data inspection, filtering, grouping, aggregation, descriptive statistical analysis, and correlation analysis using Pandas and visualize the results using Matplotlib.

## ALGORITHM 1

Dataset Inspection

1. Import dataset
2. Display dataset
3. Check dimensions
4. Display data types
5. Generate summary

## PROGRAM 1

```
import pandas as pd
df=pd.read_csv(r"C:\Users\Keertu\Downloads\tourism_dataset_5000.csv")
print(df)
```

**OUTPUT**

```
                                        Sites Visited  Tour Duration  \
0     ['Eiffel Tower', 'Great Wall of China', 'Taj M...              7
1                               ['Great Wall of China']              1
2                                     ['Eiffel Tower']              2
3                         ['Machu Picchu', 'Taj Mahal']              5
4     ['Machu Picchu', 'Taj Mahal', 'Great Wall of C...              7
...                                                 ...            ...
4995                            ['Great Wall of China']              2
4996                                      ['Colosseum']              1
4997                                      ['Taj Mahal']              1
4998  ['Taj Mahal', 'Great Wall of China', 'Colosseum']              7
4999                            ['Great Wall of China']              1

      Route ID  Tourist Rating  System Response Time  Recommendation Accuracy  \
0         1000             1.6                 3.73                       97
1         2000             2.6                 2.89                       90
2         3000             1.7                 2.22                       94
3         4000             2.0                 2.34                       92
4         5000             3.7                 2.00                       96
...        ...             ...                  ...                      ...
4995   4996000             3.9                 2.53                      100
4996   4997000             4.3                 1.53                      100
4997   4998000             3.2                 3.99                       89
4998   4999000             2.5                 2.13                       98
4999   5000000             1.9                 2.99                       92

      VR Experience Quality  Satisfaction
0                       4.5             3
1                       4.5             3
2                       4.7             3
3                       4.7             3
4                       4.8             4
...                     ...           ...
4995                    4.4             4
4996                    5.0             4
4997                    4.4             3
4998                    4.4             3
4999                    4.8             3

[5000 rows x 14 columns]
```

**PROGRAM 2**

print(df.shape)

**OUTPUT**

```
(5000, 14)
```

**PROGRAM 3:**

df.dtypes

**OUTPUT**

```
Tourist ID                   int64
Age                          int64
Interests                   object
Preferred Tour Duration      int64
Accessibility                 bool
Site Name                   object
Sites Visited               object
Tour Duration                int64
Route ID                     int64
Tourist Rating             float64
System Response Time       float64
Recommendation Accuracy      int64
VR Experience Quality      float64
Satisfaction                 int64
dtype: object
```

**ALGORITHM 2**

Filtering Data

1. Specify condition
2. Apply filter
3. Display records

**PROGRAM 4 :**

```
high_rating = df[df["Tourist Rating"] > 4]


print(high_rating)
```

**OUTPUT**

```
      Preferred Tour Duration  Accessibility           Site Name  \
6                           7          False         Eiffel Tower
13                          8           True            Colosseum
22                          8           True  Great Wall of China
28                          7           True  Great Wall of China
29                          6           True         Eiffel Tower
...                       ...            ...                  ...
4981                        5           True  Great Wall of China
4983                        3          False             Taj Mahal
4991                        6           True         Eiffel Tower
4994                        4          False         Eiffel Tower
4996                        7           True  Great Wall of China

                                    Sites Visited  Tour Duration  \
6           ['Eiffel Tower', 'Colosseum', 'Taj Mahal']              6
13                                ['Eiffel Tower']              1
22                                  ['Colosseum']              3
28    ['Taj Mahal', 'Great Wall of China', 'Eiffel T...              8
29                ['Taj Mahal', 'Great Wall of China']              2
...                                                ...            ...
4981      ['Machu Picchu', 'Eiffel Tower', 'Taj Mahal']              4
4983                        ['Colosseum', 'Taj Mahal']              2
4991                    ['Machu Picchu', 'Taj Mahal']              4
4994                    ['Taj Mahal', 'Machu Picchu']              5
4996                                ['Colosseum']              1

      Route ID  Tourist Rating  System Response Time  Recommendation Accuracy  \
6         7000             4.1                 4.66                       93
13       14000             4.3                 3.48                       91
22       23000             4.2                 1.97                       88
28       29000             4.1                 4.57                       91
29       30000             4.4                 2.18                       95
...        ...             ...                  ...                      ...
4981   4982000             4.6                 3.64                       90
4983   4984000             4.8                 3.43                       87
4991   4992000             5.0                 2.25                       90
4994   4995000             4.3                 2.98                       97
4996   4997000             4.3                 1.53                      100

      VR Experience Quality  Satisfaction
6                       4.5             4
13                      4.4             4
22                      5.0             4
28                      4.6             4
29                      4.9             4
...                     ...           ...
4981                    4.5             5
4983                    4.7             5
4991                    4.7             5
4994                    4.6             4
4996                    5.0             4

[1213 rows x 14 columns]
```

**PROGRAM 5**

**history_tourists = df[df["Interests"] == "History"]**

**print(history_tourists)**

**OUTPUT**

```
Empty DataFrame
Columns: [Tourist ID, Age, Interests, Preferred Tour Duration, Accessibility, Site Name, Sites Visited, Tour Duration, Route ID, Tourist Rating, System
Response Time, Recommendation Accuracy, VR Experience Quality, Satisfaction]
Index: []
```

**ALGORITHM 3**

Sorting Data

1. Select column
2. Apply sort_values()
3. Display sorted result

**PROGRAM 6**

```
sorted_df = df.sort_values(
    by="Satisfaction",
    ascending=False
)
```

**print(sorted_df)**

**OUTPUT**

```
      Tourist ID  Age                          Interests  \
621          622   52                    ['Architecture']
3245        3246   20                ['Architecture', 'Art']
3897        3898   39           ['History', 'Architecture']
4647        4648   48                ['Art', 'Architecture']
3504        3505   51          ['Nature', 'Cultural', 'Art']
...          ...  ...                                  ...
2049        2050   28                             ['Art']
2052        2053   36  ['History', 'Architecture', 'Nature']
2053        2054   65     ['Art', 'Architecture', 'History']
2056        2057   38                ['Architecture', 'Art']
4999        5000   32                             ['Nature']

      Preferred Tour Duration  Accessibility            Site Name  \
621                         3          False         Eiffel Tower
3245                        6           True           Colosseum
3897                        6          False  Great Wall of China
4647                        5          False           Colosseum
3504                        3           True           Taj Mahal
...                       ...            ...                 ...
2049                        7          False         Machu Picchu
2052                        5          False           Taj Mahal
2053                        6          False           Taj Mahal
2056                        4          False           Taj Mahal
4999                        4          False           Colosseum

                                  Sites Visited  Tour Duration  \
621                    ['Eiffel Tower', 'Colosseum']              5
3245                                ['Colosseum']              3
3897       ['Eiffel Tower', 'Colosseum', 'Taj Mahal']              7
4647  ['Colosseum', 'Great Wall of China', 'Machu Pi...              6
3504                                ['Taj Mahal']              3
...                                           ...            ...
2049      ['Taj Mahal', 'Eiffel Tower', 'Machu Picchu']              8
2052                                ['Taj Mahal']              1
2053                              ['Eiffel Tower']              3
2056                              ['Machu Picchu']              2
4999                        ['Great Wall of China']              1

      Route ID  Tourist Rating  System Response Time  Recommendation Accuracy  \
621     622000             4.6                  4.90                       98
3245   3246000             4.5                  2.65                       91
3897   3898000             4.5                  4.74                       93
4647   4648000             4.6                  3.35                       87
3504   3505000             4.5                  3.90                       87
...        ...             ...                   ...                      ...
2049   2050000             3.1                  2.86                       87
2052   2053000             2.7                  4.25                       90
2053   2054000             2.0                  2.81                       89
2056   2057000             2.9                  4.13                       95
4999   5000000             1.9                  2.99                       92

      VR Experience Quality  Satisfaction
621                     4.3             5
3245                    4.7             5
3897                    4.5             5
4647                    4.3             5
3504                    4.8             5
...                     ...           ...
2049                    4.7             3
2052                    4.3             3
```

**VISUALIZATION**

```
print("Mean =",
    df["Satisfaction"].mean())

print("Median =",
    df["Satisfaction"].median())

print("Mode =",
    df["Satisfaction"].mode())

print("Standard Deviation =",
    df["Satisfaction"].std())
```

**OUTPUT**

```
Mean = 3.5342
Median = 3.0
Mode = 0    3
Name: Satisfaction, dtype: int64
Standard Deviation = 0.7360290128395618
```

**ALGORITHM 4**

Statistical Analysis

1. Select numerical column
2. Calculate Mean
3. Calculate Median
4. Calculate Mode
5. Calculate Standard Deviation

**PROGRAM 7**

```
print(df.describe())
```

**OUTPUT**

```
        Tourist ID          Age  Preferred Tour Duration  Tour Duration  \
count  5000.000000  5000.000000              5000.00000     5000.00000
mean   2500.500000    43.886000                 5.52280        3.97720
std    1443.520003    15.268858                 1.71186        1.99967
min       1.000000    18.000000                 3.00000        1.00000
25%    1250.750000    30.000000                 4.00000        2.00000
50%    2500.500000    44.000000                 6.00000        4.00000
75%    3750.250000    57.000000                 7.00000        5.00000
max    5000.000000    70.000000                 8.00000        9.00000

           Route ID  Tourist Rating  System Response Time  \
count  5.000000e+03     5000.000000           5000.000000
mean   2.500500e+06        3.002740              3.238802
std    1.443520e+06        1.164353              1.016123
min    1.000000e+03        1.000000              1.500000
25%    1.250750e+06        2.000000              2.360000
50%    2.500500e+06        3.000000              3.230000
75%    3.750250e+06        4.000000              4.130000
max    5.000000e+06        5.000000              5.000000

       Recommendation Accuracy  VR Experience Quality  Satisfaction
count              5000.000000            5000.000000   5000.000000
mean                 92.492200               4.493160      3.534200
std                   4.616081               0.290596      0.736029
min                  85.000000               4.000000      3.000000
25%                  88.000000               4.200000      3.000000
50%                  92.000000               4.500000      3.000000
75%                  97.000000               4.700000      4.000000
max                 100.000000               5.000000      5.000000
```

**PROGRAM 8**

```
accessibility_avg = df.groupby(
    "Accessibility"
)["Tourist Rating"].mean()


print(accessibility_avg)
```

**OUTPUT**

```
Accessibility
False    3.006914
True     2.998461
Name: Tourist Rating, dtype: float64
```

**ALGORITHM 5**

Grouping and Aggregation

1. Select Department Column
2. Apply groupby()
3. Calculate Average Marks
4. Display Result

**PROGRAM 9**

```
corr = df[[
    "Recommendation Accuracy",
```

**"Satisfaction"**

**]].corr()**

**print(corr)**

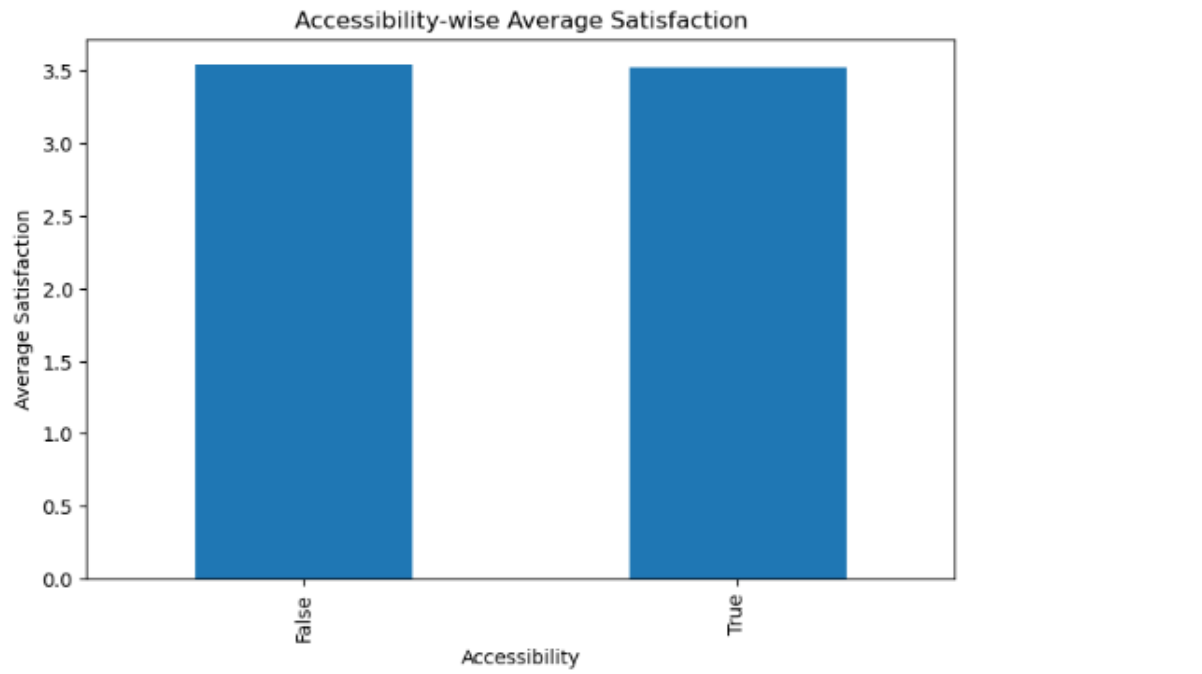
**OUTPUT**

| | Recommendation Accuracy | Satisfaction |
|---|---|---|
| Recommendation Accuracy | 1.000000 | -0.002453 |
| Satisfaction | -0.002453 | 1.000000 |

**VISUALIZATION**

```
import matplotlib.pyplot as plt


access_avg = df.groupby("Accessibility")["Satisfaction"].mean()


access_avg.plot(kind="bar", figsize=(8,5))


plt.xlabel("Accessibility")

plt.ylabel("Average Satisfaction")

plt.title("Accessibility-wise Average Satisfaction")


plt.show()
```

**OUTPUT**




**RESULT**

Thus, data inspection, filtering, sorting, descriptive statistical analysis, grouping, aggregation, and correlation analysis were successfully performed using Pandas. The analytical results were visualized and interpreted successfully.